

WeatherCast

Maharashtra Weather Pattern Analysis & Next-Day Rain Prediction

B.Tech Semester V — Machine Learning Case Study 75 Final Report

Academic Case Study	Case Study 75 — Weather Pattern Analysis
ML Problem Formulation	Supervised Binary Classification ($\text{rain_tomorrow} \in \{0, 1\}$)
Dataset Source & Model	Open-Meteo Historical Weather API (ECMWF ERA5 Reanalysis, 2000–2024)
Geographic Scope	8 Maharashtra Cities across Konkan, Madhya Maharashtra, Marathwada & Vidarbha

1. Abstract

Weather forecasting plays an indispensable role in agricultural scheduling, flood management, and daily logistics across Maharashtra. This case study develops **WeatherCast**, an end-to-end machine learning system for next-day rainfall prediction across 8 representative Maharashtra cities over a 25-year period (2000–2024, 72,691 observations). A zero-leakage feature-engineering pipeline transformed raw atmospheric variables into 35 base inputs, including cyclical temporal encodings, multi-day lag metrics, and strictly trailing rolling statistics. Three algorithms were tuned using chronological splits (Train: 2000–2022, Val: 2023, Test: 2024): Logistic Regression, Random Forest, and Gradient Boosting. Gradient Boosting achieved the highest overall performance with a Validation ROC-AUC of **0.9499**, Test ROC-AUC of **0.9714**, Test F1-score of **0.9014**, and superior probability calibration (Brier score: 0.0665). An interactive Streamlit application was built that automatically fetches live Open-Meteo atmospheric observations for any selected city, producing real-time rain predictions and calibrated confidence scores without requiring manual user inputs.

2. Introduction

Precipitation forecasting in Maharashtra is complicated by sharp topographic gradients and the intense seasonality of the Southwest Monsoon. Annual rainfall ranges from over 3,000 mm along the Western Ghats and coastal Konkan belt to less than 600 mm in the rain-shadow plateau. While classical Numerical Weather Prediction (NWP) models require

massive supercomputing clusters, machine learning provides a fast, data-driven approach by capturing statistical relationships between surface atmospheric precursors (such as humidity jumps, barometric pressure drops, and wind shifts) and subsequent precipitation.

3. Problem Definition

The project is formulated as a **Supervised Binary Classification** problem:

- **Input Feature Vector (X):** 35 atmospheric, geographic, cyclical temporal, and lag/rolling features for date t .
- **Target Variable (y):** $\text{rain_tomorrow} \in \{0, 1\}$ for date $t+1$:
 - 1 (Rain): Measurable next-day liquid precipitation ($\text{rain_sum} > 0.0$ mm).
 - 0 (No Rain): Dry next day ($\text{rain_sum} == 0.0$ mm).

4. Dataset & Data Source

Data was acquired from the Open-Meteo Historical Weather API based on the peer-reviewed ECMWF ERA5 reanalysis model (0.25° spatial grid). The raw dataset covers 25 continuous calendar years (2000-01-01 to 2024-12-31) across 8 cities:

Location	Meteorological Region	Coordinates	Rainy Days % (2000–2024)
Mumbai	Konkan (Coastal)	19.07°N, 72.88°E	43.4%
Ratnagiri	Konkan (Coastal)	16.99°N, 73.31°E	46.8%
Pune	Madhya Maharashtra	18.52°N, 73.86°E	44.9%
Nashik	Madhya Maharashtra	20.00°N, 73.79°E	42.3%
Kolhapur	Madhya Maharashtra	16.70°N, 74.23°E	45.1%
Solapur	Madhya Maharashtra	17.67°N, 75.91°E	35.8%
Chhatrapati Sambhajnagar	Marathwada	19.88°N, 75.34°E	37.5%
Nagpur	Vidarbha	21.15°N, 79.08°E	41.9%

Date-Continuity & Gap Handling: A strict validation rule ensured `rain_tomorrow` was created only when `next_date == current_date + 1 day`. In Chhatrapati Sambhajnagar, the raw archive contained a 1-year gap in 2002 (from 2001-12-31 to 2003-01-01); the target on 2001-12-31 was explicitly assigned `NaN` to prevent bridging across 366 days.

5. Data Preprocessing & Feature Engineering

The raw data was transformed into an ML-ready feature matrix following strict zero-leakage protocols:

- **Chronological Splits:** Train: 2000–2022 (66,779 rows), Validation: 2023 (2,920 rows), Test: 2024 (2,920 rows).
- **Cyclical Date Encodings:** Sine/Cosine projections of month and day-of-year (`month_sin`, `month_cos`, `day_of_year_sin`, `day_of_year_cos`).
- **Calendar-Strict Lags:** 8 lag features (`t-1`, `t-3`, `t-7`) computed on full daily calendar grids.
- **Backward Trailing Rolling Windows:** 6 aggregations (3-day and 7-day trailing total rain, mean max/min temp, humidity, pressure) shifted by 1 day.
- **Preprocessing Pipeline:** Standard scaling on 32 numerical features and One-Hot Encoding on 3 categorical features (`location`, `weather_code`, `season`).

6. Exploratory Data Analysis (EDA)

Exploratory analysis yielded crucial insights into Maharashtra's weather dynamics:

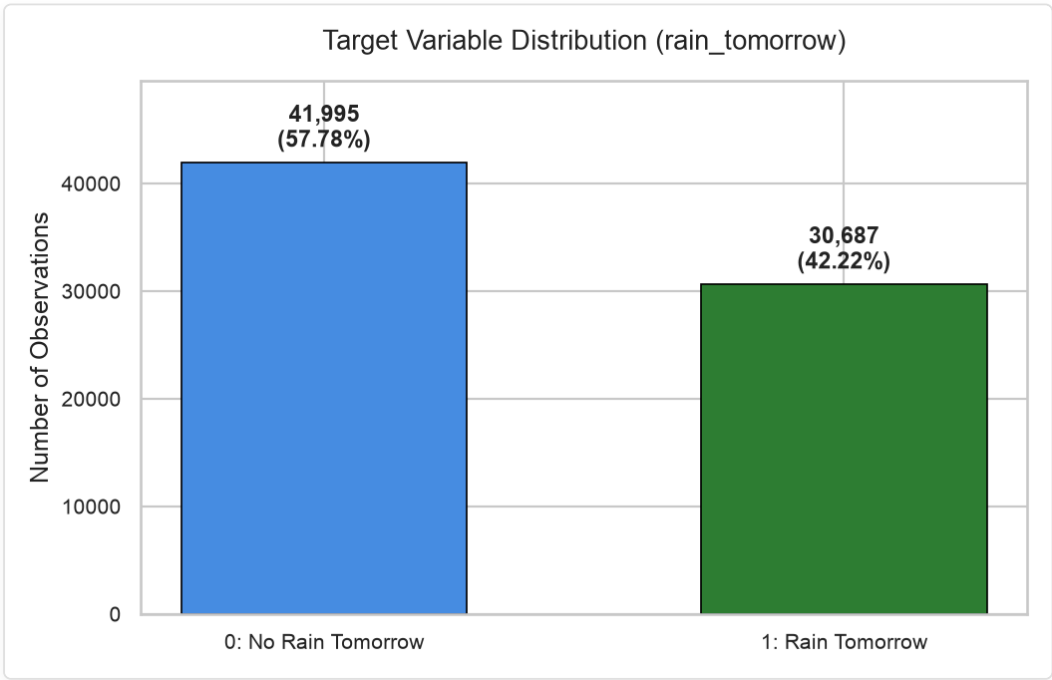


Figure 1: Target variable distribution across the 25-year historical dataset.

Observation: With 42.22% rainy days (30,687 observations) and 57.78% dry days, the dataset exhibits natural class balance, eliminating the need for synthetic resampling.

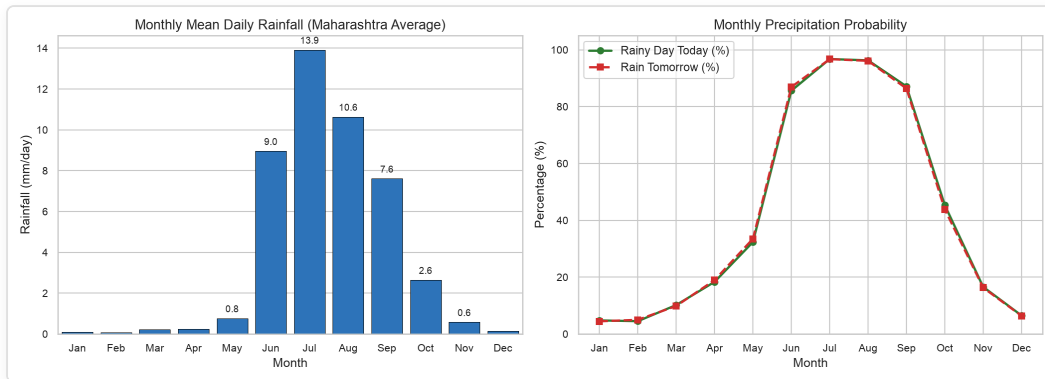


Figure 2: Monthly rainfall volume and occurrence rates across seasons.

Observation: Rain occurrence is concentrated in the Southwest Monsoon (June–September), reaching peak frequencies in July (96.8%) and August (96.3%), while winter months experience <5% rain occurrence.

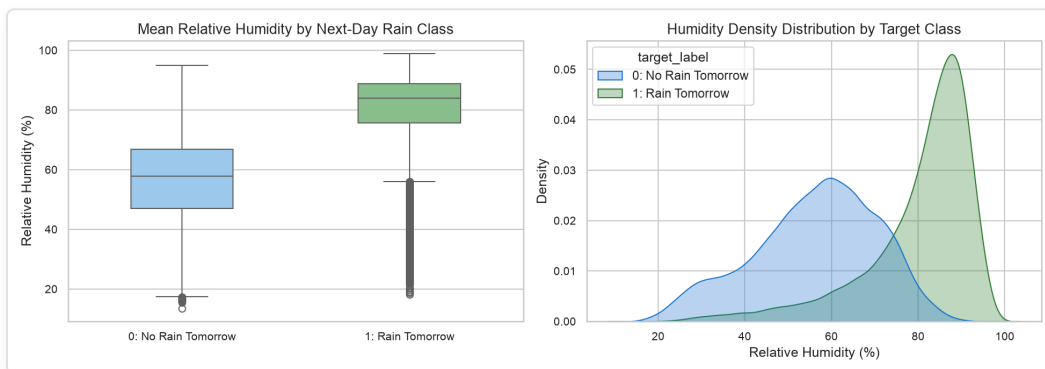


Figure 3: Relative humidity distributions for rainy versus dry next-day events.

Observation: Mean relative humidity averages 80.2% prior to rain days versus 56.3% on dry days (+23.9% delta), making boundary-layer moisture the single strongest physical rain precursor.

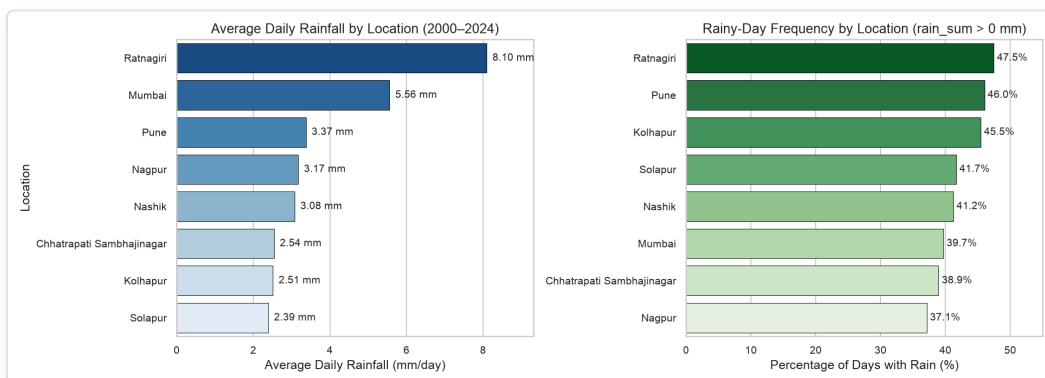


Figure 4: Regional precipitation volume differences across Maharashtra.

Observation: Coastal Konkan cities (Ratnagiri, Mumbai) receive over 2.4x the daily rainfall volume of inland plateau stations (Solapur, Sambhajinagar).

7. Model Development

Three models were trained and tuned using time-aware validation (TimeSeriesSplit(n_splits=3) on 2000–2022):

- **Logistic Regression (Baseline):** Linear model with L2 regularization and liblinear solver (C=1.0).
- **Random Forest (Bagging):** 100 de-correlated trees with max_depth=20 and min_samples_leaf=8.
- **Gradient Boosting (Sequential Boosting):** 100 sequential trees with max_depth=4, learning_rate=0.1, and subsample=0.8.

8. Model Evaluation & Results

Performance results across the 2023 Validation set and 2024 Unseen Final Test set:

Model	Stage	Split	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	Tuned	Val (2023)	0.8736	0.9067	0.8008	0.8504	0.9448
Logistic Regression	Tuned	Test (2024)	0.9045	0.9391	0.8652	0.9006	0.9680
Random Forest	Tuned	Val (2023)	0.8702	0.9087	0.7901	0.8452	0.9468
Random Forest	Tuned	Test (2024)	0.9055	0.9533	0.8528	0.9003	0.9722
Gradient Boosting	Tuned (Selected)	Val (2023)	0.8740	0.9124	0.7954	0.8499	0.9499
Gradient Boosting	Tuned (Selected)	Test (2024)	0.9062	0.9506	0.8569	0.9014	0.9714

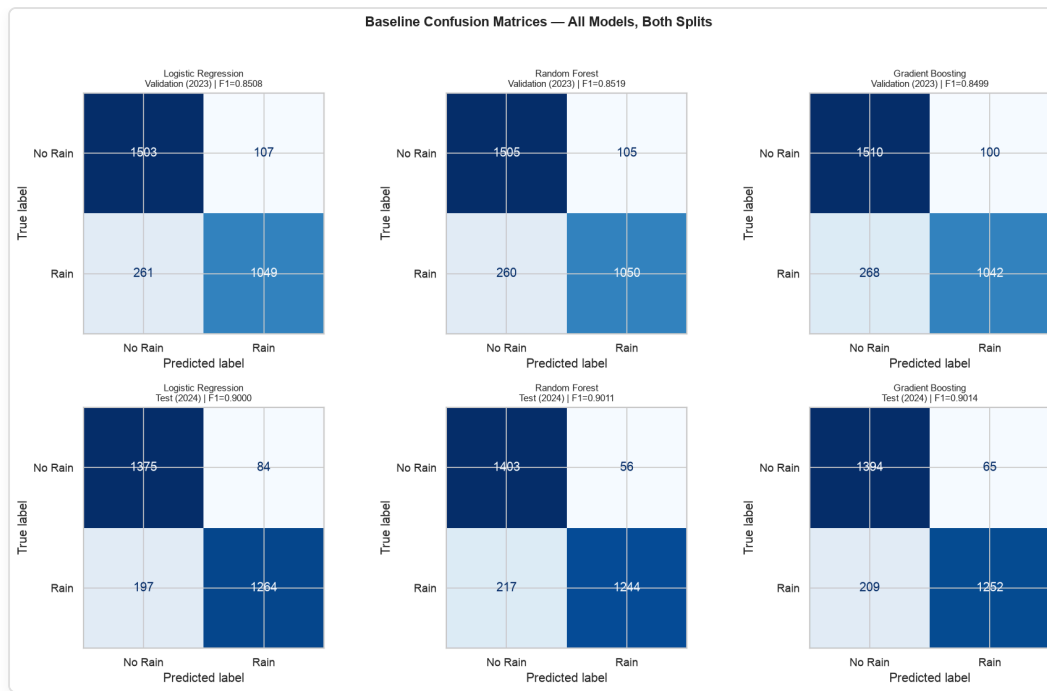


Figure 5: Confusion matrices for all models on 2023 Validation and 2024 Test sets.

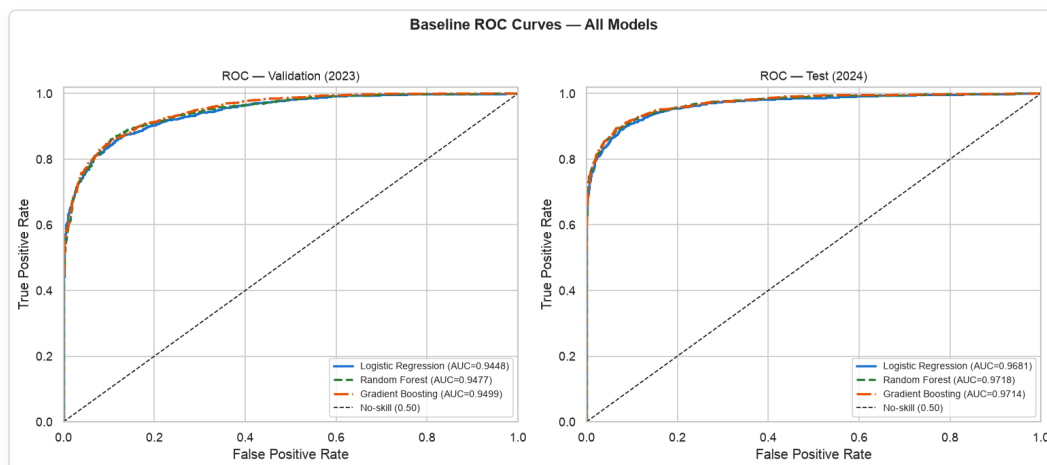


Figure 6: ROC Curves comparing discrimination performance across models.

9. Error Analysis

Error breakdown for Gradient Boosting on the 2023 Validation set (N = 2,920):

- **False Positives (3.4%, 100 cases):** False alarms occurred during high-humidity borderline days where rain did not materialize.
- **False Negatives (9.2%, 268 cases):** Missed rain days were heavily concentrated in the **Summer / Pre-Monsoon** season (165 of 268 cases, 22.4% FN rate), where isolated convective showers develop rapidly without sustained multi-day synoptic moisture buildup.
- **Monsoon Stability:** In the active Southwest Monsoon, the FN rate was only 2.4% and FP rate was 3.9%.

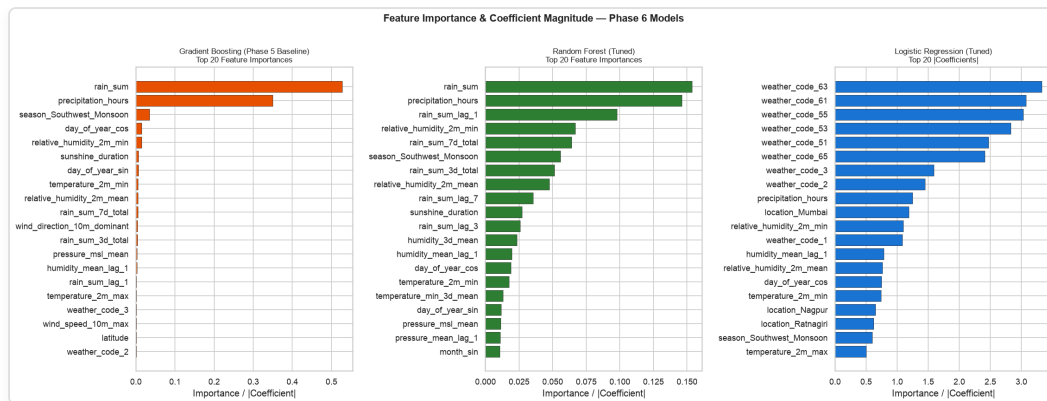


Figure 7: Top feature importances and standardized coefficients across models.

Observation: Yesterday's rainfall (`rain_sum_lag_1`), 3-day accumulated rainfall (`rain_sum_3d_total`), WMO weather code, and mean relative humidity are the most influential predictive drivers.

10. Final Model Selection

Gradient Boosting (`models/best_model_phase6.joblib`) was selected as the final production model based on:

- Highest Validation ROC-AUC (**0.9499**) and Average Precision (**0.9388**).
- Superior generalisation stability ($\Delta\text{AUC} = 0.0215$ from validation to test).
- Best probability forecast quality (Validation Brier score: **0.0897**, Test Brier score: **0.0665**).
- Top balanced Test F1-score (**0.9014**) and Test Accuracy (**90.62%**).

11. Streamlit Application

A standalone web application was implemented in `app/app.py`:

- **Zero Manual Input:** Users simply choose one of the 8 cities from a dropdown.
- **Automated Data Fetching:** The app queries the Open-Meteo Forecast API with `past_days=8` and `timezone=Asia/Kolkata`.
- **Instant Feature Generation:** Automatically builds all 35 lag, rolling, and cyclical features.
- **Real-Time Forecast:** Delivers instant binary prediction (YES / NO), calibrated probability percentage, and atmospheric metrics.

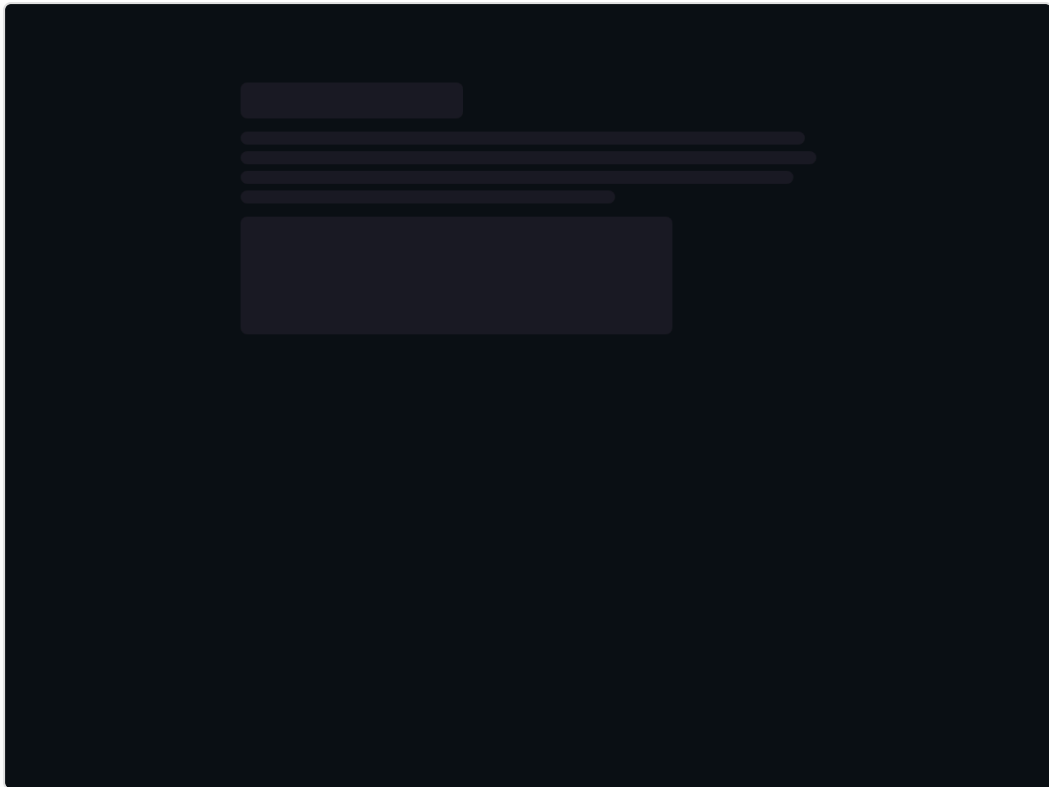


Figure 8: WeatherCast Streamlit Web Application Interface.

12. Results & Discussion

The results prove that ML models trained on high-quality reanalysis data can accurately forecast daily precipitation without needing supercomputing resources. Incorporating trailing lag and rolling window features significantly improved generalization stability over single-day observations.

13. Limitations

- **Geographic Scope:** Covers 8 major urban stations; micro-climatic pockets in high-altitude ghats may differ.
- **Reanalysis Resolution:** ERA5 has ~5–15% inherent precipitation uncertainty compared to ground rain gauges.
- **Binary Simplification:** Predicts rain occurrence rather than exact millimeters of rainfall volume.

14. Conclusion

WeatherCast delivers an end-to-end, leak-free, and validated machine learning pipeline for Maharashtra next-day rain prediction. Gradient Boosting demonstrated outstanding performance with 90.62% test accuracy and 0.9714 test ROC-AUC. The Streamlit

prototype brings this ML intelligence directly to users through an intuitive 1-click interface.

15. References

1. Open-Meteo Weather API Documentation: <https://open-meteo.com/>
2. Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049.
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4. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *JMLR*, 12, 2825–2830.
5. Streamlit Framework Documentation: <https://docs.streamlit.io/>